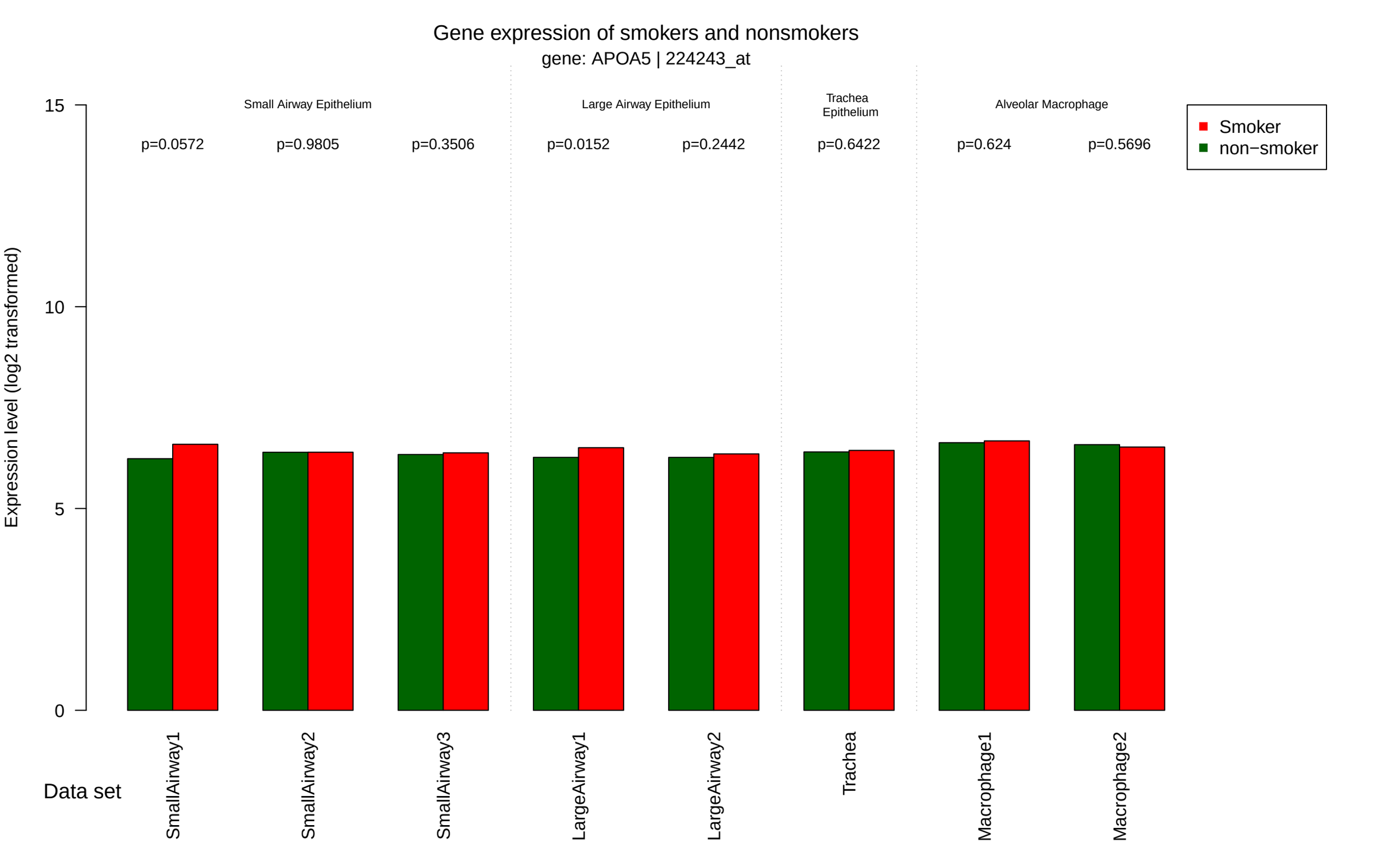
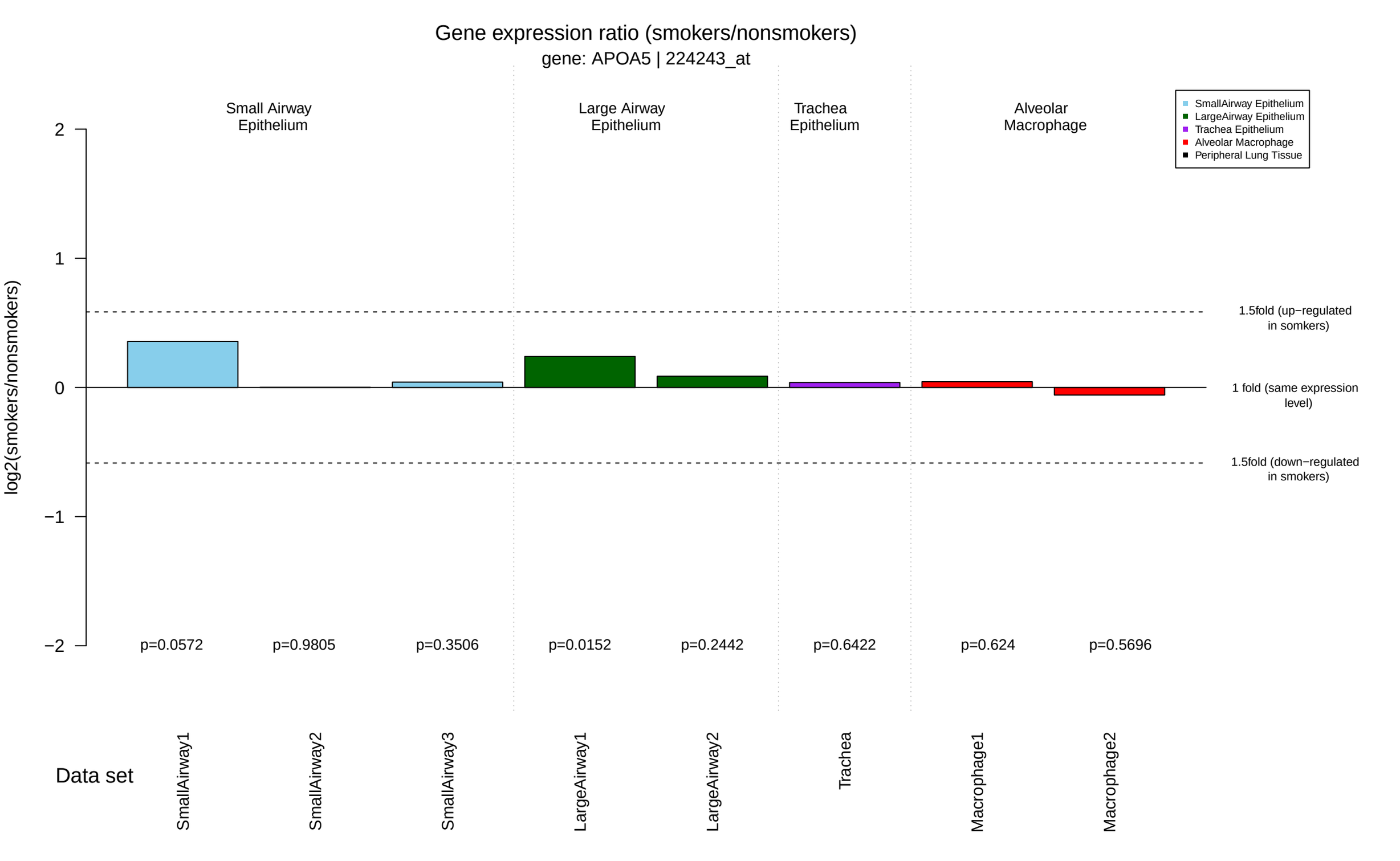
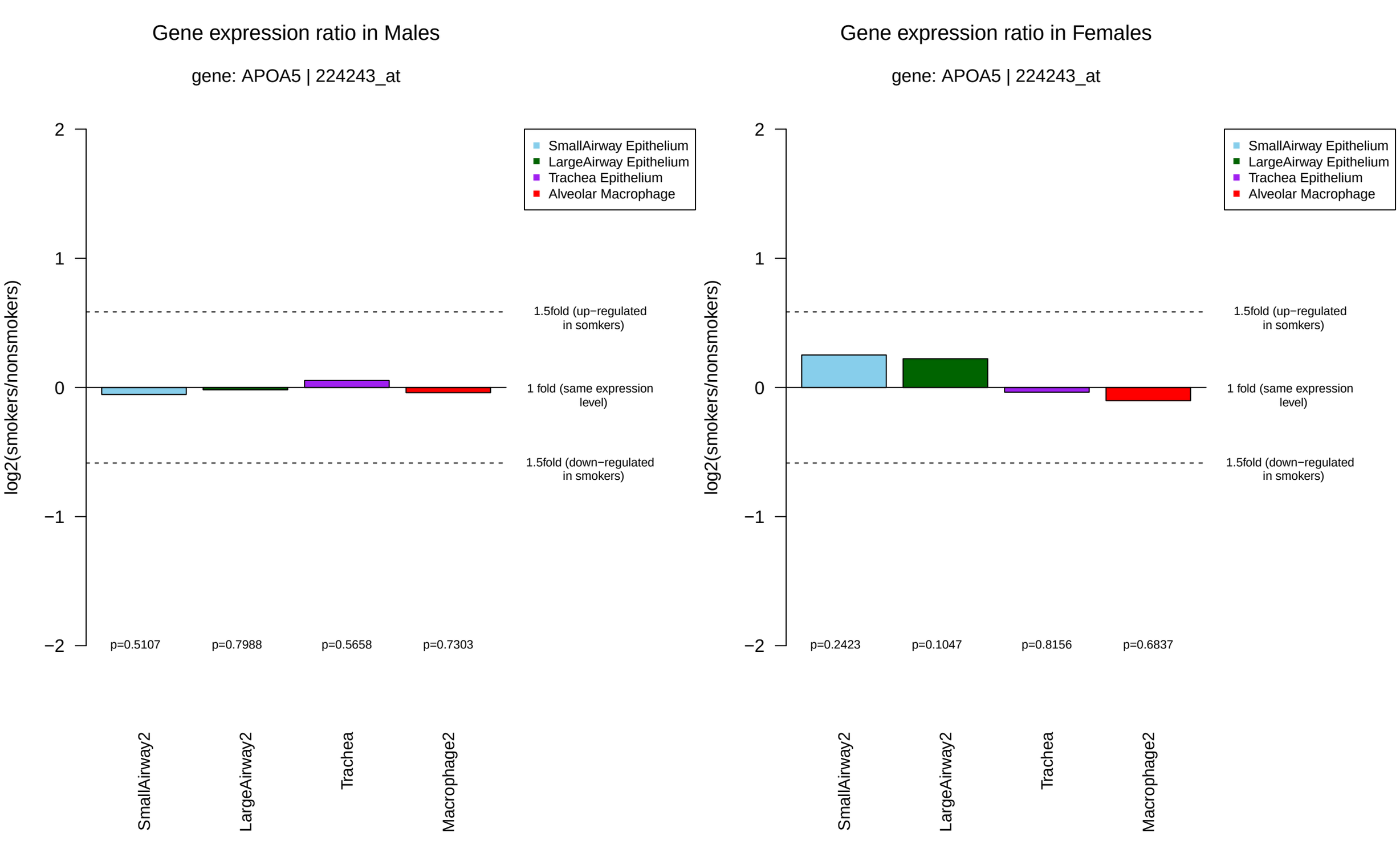
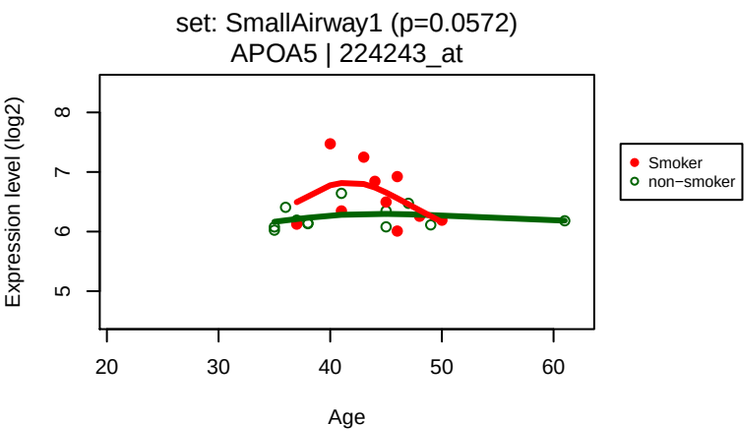
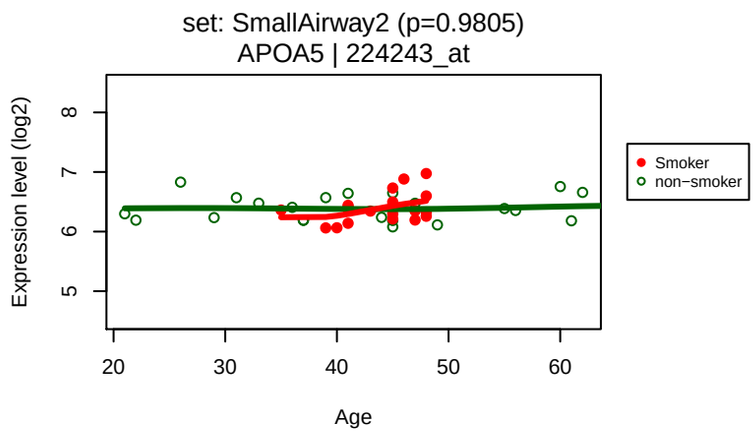
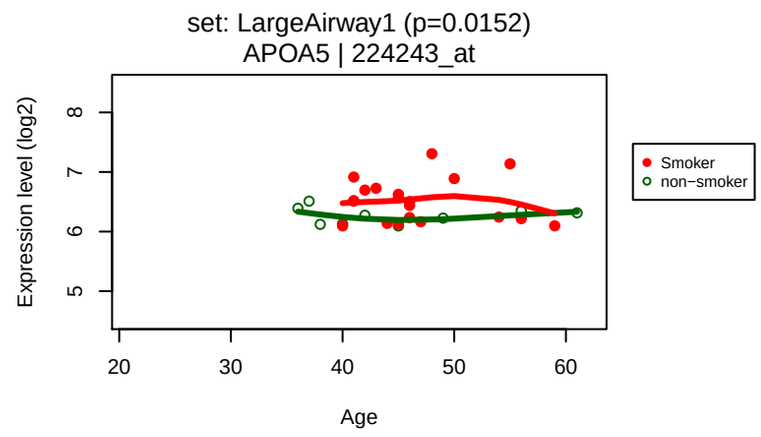

Gene expression of smokers and nonsmokers
gene: APOA5 | 224243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0572
p=0.9805
p=0.3506
p=0.0152
p=0.2442
p=0.6422
p=0.624
p=0.5696
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

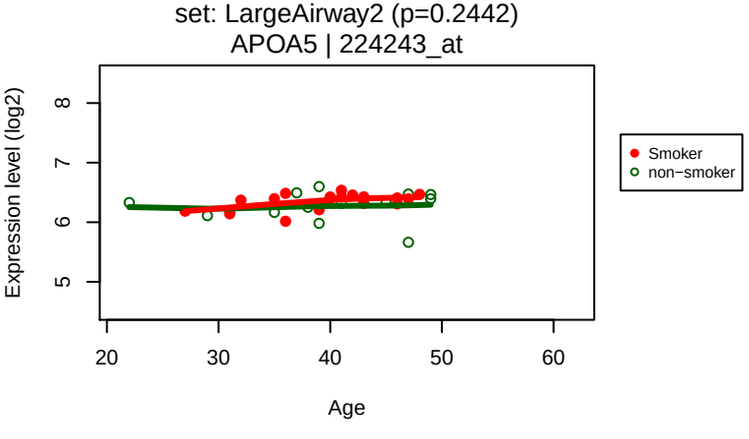

Gene expression ratio (smokers/nonsmokers)
gene: APOA5 | 224243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0572
p=0.9805
p=0.3506
p=0.0152
p=0.2442
p=0.6422
p=0.624
p=0.5696
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

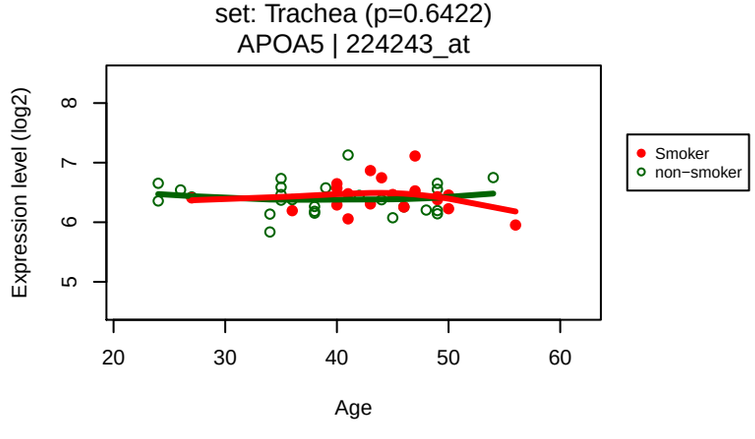

Gene expression ratio in Males
gene: APOA5 | 224243_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5107
p=0.7988
p=0.5658
p=0.7303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOA5 | 224243_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2423
p=0.1047
p=0.8156
p=0.6837
SmallAirway2
LargeAirway2
Trachea
Macrophage2

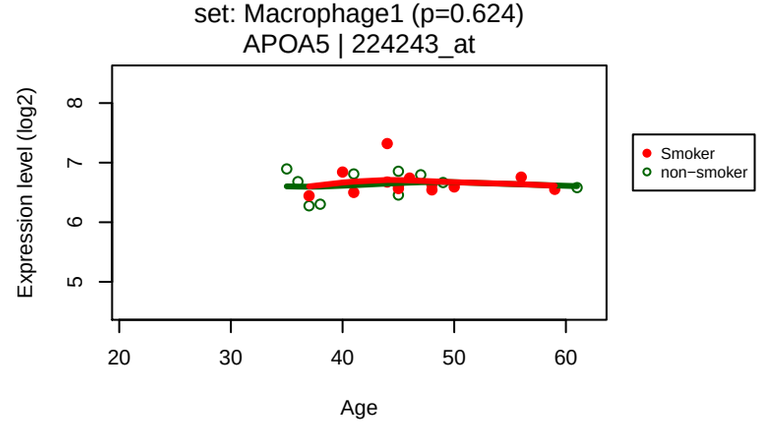

set: SmallAirway1 (p=0.0572)
APOA5 | 224243_at
Expression level (log2)
Age
Smoker
non−smoker

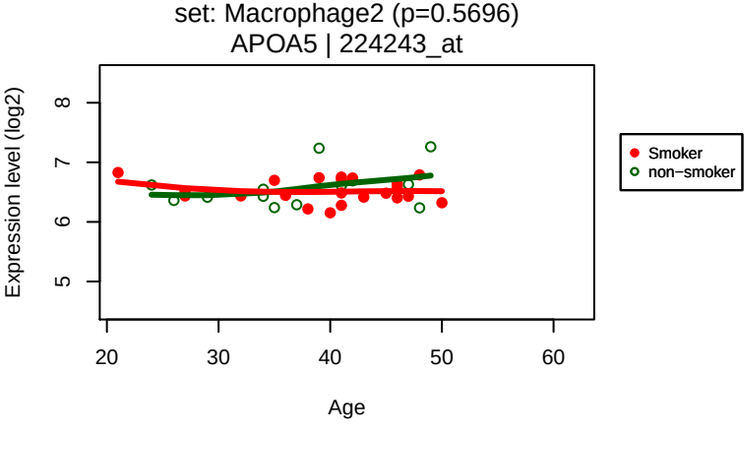

set: SmallAirway2 (p=0.9805)
APOA5 | 224243_at
Expression level (log2)
Age
Smoker
non−smoker

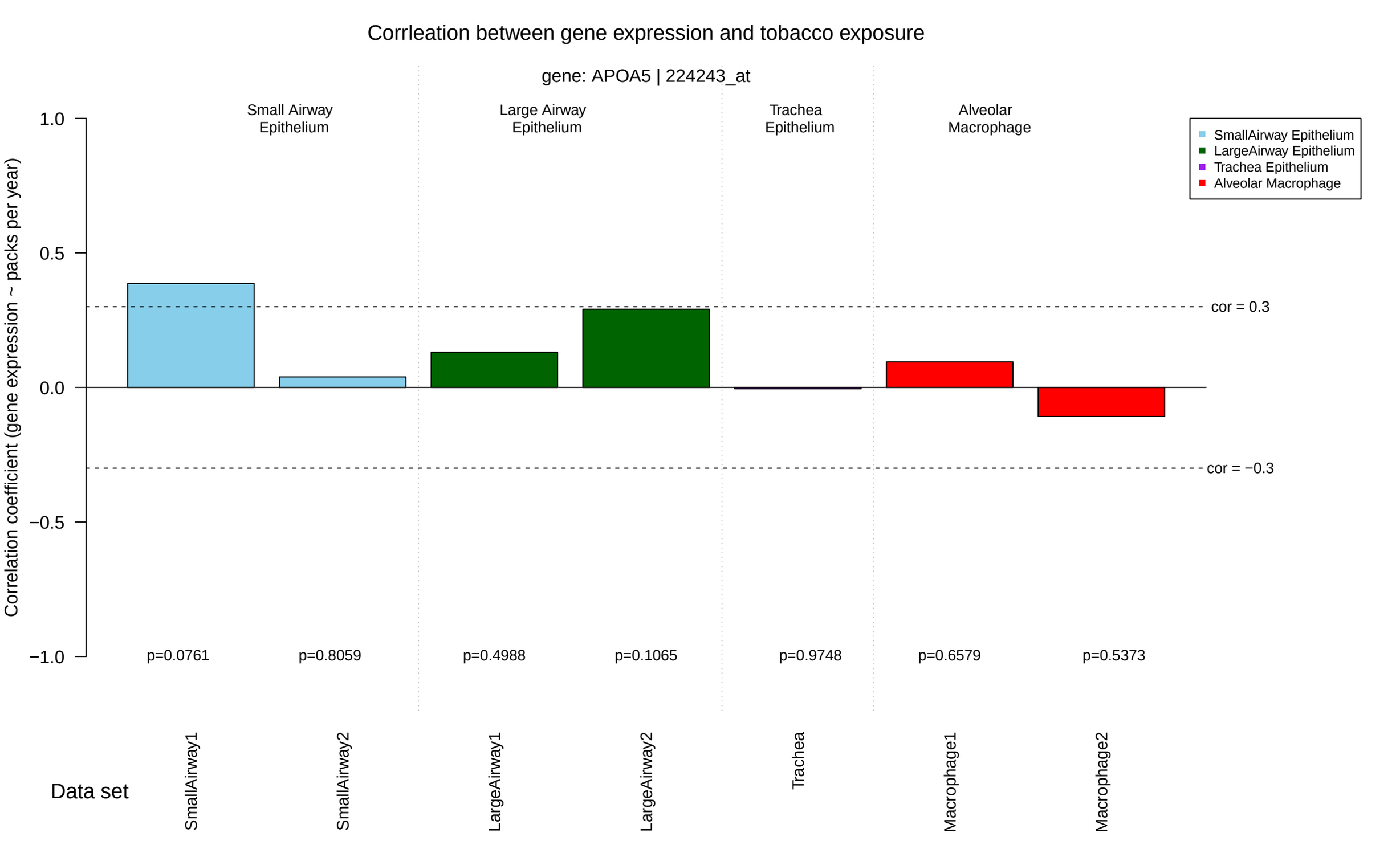

set: LargeAirway1 (p=0.0152)
APOA5 | 224243_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2442)
APOA5 | 224243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6422)
APOA5 | 224243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.624)
APOA5 | 224243_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5696)
APOA5 | 224243_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOA5 | 224243_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0761
p=0.8059
p=0.4988
p=0.1065
p=0.9748
p=0.6579
p=0.5373
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2